Lesson plan · for the teacher

Teaching decisions: unit check

Optimization · Level 4 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Two goals at once
- Fair by design
- When the model is wrong
- Big problems, good answers

Activity: Decision clinic

1. Warm-up (5 min): Each teacher names one school decision that people complain about: the timetable, duties, exam week or the bus.
2. Quiz (15 min): Teams of four read each question aloud, agree one answer, then hear the explanation. Write the hardest question on the board.
3. Clinic (15 min): Each team takes one decision from the warm-up and fills four boxes: the options nobody beats, the fairness rule, the missing rules people would spot, and what good enough means.
4. Share (5 min): Each team names one change it will propose to the head teacher, and who must agree it.

Works as a staff training session as well as a class review: use real school decisions, and keep names out of the examples.

Discussion

- Which idea from this unit will you use first?
- Where does our school hide a trade-off inside a single number?
- Who should be in the room when a fairness rule is chosen?

Exit ticket

- One decision I will lead differently, and the first step I will take.

Try it in class tomorrow

Pick one decision at your school this month and run it the way this unit suggests: cross out the beaten options, agree the weights and a fairness rule before anyone sees the winner, test the plan with the people it affects, and publish how it was made.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Three bus routes: X (30 km, longest ride 50 min), Y (32 km, 45 min) and Z (33 km, 50 min). Which route is beaten?**

◯ X ◯ Y ◯ Z ◯ None of them

2 **The weights for combining two goals are facts that the person building the model should work out alone.**

◯ True ◯ False

3 **Which of these are signs that a model is missing a rule? Choose all that apply.**

☐ The cooks say the plan can't be cooked in time
☐ The cheapest lunch serves a spoonful of beans
☐ Last year's real timetable breaks one of the model's rules
☐ The solver reports OPTIMAL
☐ The plan barely changes when a price changes

4 **Match each idea to what it means.**

Pareto front	• • • •	A plan no single small change improves
Shadow price	• • • •	What the total gives up for the worst-off
Price of fairness	• • • •	What one more unit of a limit is worth
Local best	• • • •	The plans no other plan beats

5 **What does this program print?**

```
plans = [("A", 4, 9), ("B", 5, 4), ("D", 6, 1)]
w = 4  # one exam day counts as 4 pupils
best = min(plans, key=lambda p: w * p[1] + p[2])
print(best[0])
```

◯ A ◯ B ◯ D